Prediction Is Not Identification

A minimal benchmark for choosing the experiment that matters

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Abstract

Observational prediction can be perfect while mechanism identification remains impossible. We introduce **IDENT**, a one-shot benchmark where two or more hidden mechanisms are indistinguishable under a supplied observation set, yet at least one permitted intervention separates them. We prove a finite passive-identification impossibility bound, ship a deterministic generator with exact separator annotations across three symbolic domains, and evaluate trivial baselines plus frontier models under a frozen one-intervention protocol. Construction gates pass. An oracle weakest identifying separator reaches 100% final identification. Strong models usually intervene rather than bluff, but final identification remains far below oracle—even when separator choice is perfect—revealing a residual failure in update-after-evidence.

1. Claim and scope

Operational claim. There exist tasks on which a passive learner cannot exceed chance at naming the true mechanism, while a single cheap intervention identifies it exactly.

IDENT is intentionally narrow. It is not a general theory of intelligence, a transformer training run, or a natural-language science suite. The contribution succeeds only if three pieces land together:

1. a formal impossibility result for passive learners on the benchmark family;
2. a compact generator with known latent mechanisms and exact separators;
3. an evaluation showing whether models recognize underdetermination and choose (then use) the right experiment.

2. Minimal theory

2.1 Objects

- Finite hypothesis set $H=\{h_1,\ldots,h_k\}$.
- Experiment set G with costs $c(g)\ge 0$.
- Response function $R(h,g)$.
- Initial passive family $G_0\subset G$.

2.2 Experiment-relative equivalence

For $A\subseteq G$,

$$ h_i\sim_A h_j \iff R(h_i,g)=R(h_j,g)\ \text{for all }g\in A. $$

The live class $S=[h^\star]_{\sim_{G_0}}$ is the set of mechanisms still compatible with all passive observations.

2.3 Weakest identifying separator

Intervention g **separates** S when outcomes $\{R(h,g):h\in S\}$ are not unique. It **identifies** $h^\star$ when observing $R(h^\star,g)$ leaves a singleton live class. IDENT annotates minimum-cost identifying separators.

2.4 Passive identification impossibility

Theorem (benchmark form). If $|S|=m\ge 2$ and the held-out label is non-constant on S, then no learner receiving only the passive transcript can exceed worst-case accuracy $1/m$ at recovering the true hypothesis.

Proof sketch. Any two members of S induce identical passive transcripts. A deterministic map returns one answer on that transcript and is therefore wrong on at least one world. A randomized map has minimax accuracy at most $1/m$ against an adversary choosing the true member of S. □

Active sufficiency. IDENT v1 generates only items with at least one one-step identifying separator, so failure is not caused by an impossible task.

3. Benchmark

3.1 Task interface

Each item supplies prior observations, a hypothesis menu, and candidate interventions with costs. The model may answer immediately or spend exactly one intervention, receive the outcome, then answer.

3.2 Domains

Domain	Latent object	Passive masking
`boolean_causal`	two-input Boolean mechanisms	agree on an observed input support
`finite_state`	tiny acceptors over $\{0,1\}^{\le 3}$	agree on observed traces
`small_programs`	integer functions	agree on supplied test inputs

Every item includes a wasteful “more of the same” distractor that cannot separate the live class.

3.3 Dataset

Deterministic seed `20260727` yields 1000 items: train 700 / dev 150 / test 150. Validators enforce live-class size ≥ 2, existence of an identifying separator, exact minimum-separator annotations, passive chance bounds, and basic leak checks.

3.4 Metrics

Separator accuracy, weakness regret, false-certainty rate, post-intervention identification, final accuracy, and efficiency (identification gain / cost).

4. Pre-registered gates

Gate	Result
G1 Formal ambiguity	PASS
G2 Separability / active sufficiency	PASS
G3 Passive bound (answer-now ≤ chance + Hoeffding slack)	PASS
G4 Oracle solvability (≥99% final ID)	PASS (100%)
G5 Capability gap vs oracle	PASS
G6 EIG nontriviality	PASS
G7 Label/order robustness	PASS

5. Results

5.1 Baselines (full test, n=150)

Baseline	Separator	False certainty	Final acc
answer_now	0.00	1.00	0.50

random_intervention	0.55	0.00	0.69
max_output_variance	1.00	0.00	0.92
expected_information_gain	1.00	0.00	0.93
oracle_weakest_separator	1.00	0.00	1.00

Answer-now is maximally falsely certain and remains near chance. Exact information gain already selects separators on this symbolic menu; IDENT is therefore a measurement instrument, not a claim of a new IG principle.

5.2 Frontier models via OpenRouter (test slice, n=40)

Frozen protocol; temperature 0; one permitted intervention.

System	Separator	False certainty	Post-intervention ID	Final acc
oracle	1.00	0.00	1.00	1.00
anthropic/claude-sonnet-4	1.00	0.05	0.58	0.55
openai/gpt-4o-mini	0.74	0.03	0.41	0.45
google/gemini-2.5-flash	0.80	0.00	0.25	0.23

G7 reshuffle (n=15) leaves separator accuracy within 0.07 for Claude and GPT-4o-mini.

5.3 Frontier scale-up (direct APIs, n=40)

Same frozen protocol via Doppler `cofounder/stg_superoptimizers` (`openai:gpt-5.6-sol`, `anthropic:claude-opus-5`).

System	Separator	False certainty	Post-intervention ID	Final acc
oracle	1.00	0.00	1.00	1.00
openai/gpt-5.6-sol	1.00	0.00	0.55	0.55
anthropic/claude-opus-5	1.00	0.00	0.58	0.58

Frontier models close the separator-choice gap (both 1.00, near-zero weakness regret) but **do not** close final identification. Result is essentially unchanged from Claude Sonnet 4 on final ID. G5 still passes once final/post-intervention accuracy is treated as a first-class gap (not only separator accuracy).

5.4 Failure shape

Models usually intervene. At mid-tier strength the gap mixes separator errors and update failures; at frontier strength the residual collapses to **update-after-evidence**: perfect or near-perfect separator choice with final identification stuck near 0.55–0.58. Expanding surface complexity is not required to preserve a publishable gap.

6. What this does not show

- That experiment-relative equivalence is a new concept.
- That information gain is a new learning principle.
- That model failure proves absence of "understanding."
- That the result transports to natural-language science wrappers without further leak tests.

7. Reproduce

```
python3 -m experiments.ident.experiment
python3 -m pytest tests/test_ident.py -q
doppler run --project cofounder --config dev -- \
  python3 -m experiments.ident.eval.run_models --limit 40 --no-robustness
```

Package: `experiments/ident/`. Paper source: `papers/ident/paper.md`.

8. Conclusion

IDENT makes one missing capability hard to ignore: sounding certain—or even requesting an experiment—is not the same as identifying a mechanism. Passive prediction is provably bounded on the live equivalence class; the correct one-step separator is exactly known; and current strong models leave a reproducible gap between intervention and identification.